

- 4) A bag contains 6 white and 4 red balls. One ball is drawn at random. What is the probability that it will be white?

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- 5) 15 identical balls are distributed at random into 4 boxes numbered 1, 2, 3, 4. Find the probability that, (a) each box contains at least 2 objects and (b) no box is empty.

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- 6) A box contains 20 identical tickets, the tickets being numbered as 1,2,3,..., 20. If 3 tickets are chosen at random, what is the probability that the numbers on the drawn tickets will be in arithmetic progression?

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25.5 THEOREMS OF PROBABILITY

In this section, we will consider the basic theorems of probability. We will denote the sub-sample spaces by A,B,C ... which represent the elements of the sample space S, which are favourable to the events A, B and C.

Note that $P(S)=1$ and $P(A),P(B),P(C),\dots$ lie between 0 and 1.

25.5.1 Theorem of Total Probability

If two events A and B are mutually exclusive, exhaustive and equally likely, then the probability of occurrence of either A or B, is given by $P(A + B)$ or $P(A \cup B)$ is given by the sum of their probability. Thus,

$$P(A \cup B) = P(A) + P(B)$$

This is also known as the Addition Theorem.

Let us assume that a random experiment has n possible outcomes which are mutually exclusive, exhaustive and equally likely. While m_1 of them are favourable to A, m_2 are favourable to B. By the classical definition of probability

$$P(A) = m_1/n \text{ and } P(B) = m_2/n$$

Since A and B are mutually exclusive and exhaustive, the number of m_1 cases favourable to the event A is completely distinct from m_2 cases favourable to the event B. Thus, the number of cases favourable to A and B are (AUB), is therefore given by m_1+m_2 ,

$$P(A+B) = P(A \cup B) = (m_1+m_2)/n = (m_1/n) + (m_2/n) = P(A) + P(B)$$

Deductions from Theorem of Total Probability

1) Theorem of Complementary Event

If A denotes the occurrence of the event A, then A^c (read as ‘compliment of A’) denotes non-occurrence of the event A and

$$P(A) = 1 - P(A^c).$$

Since A and A^c are mutually exclusive and exhaustive events, $S = \{A, A^c\}$. Applying the theorem of total probability we get,

$$P(S) = P(A) + P(A^c) = 1$$

$$\text{or, } P(A^c) = 1 - P(A)$$

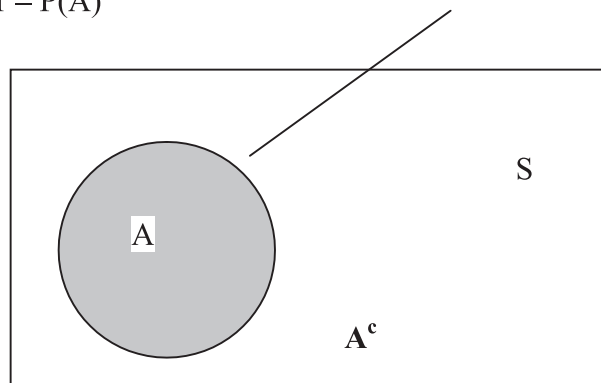


Fig. 25.2: Complementary Event

See that this theorem is very intuitive. If the probability of getting head while tossing an unbiased coin is 5, then the probability of getting a tail is obviously $0.5(1-.5)$.

2) Extension of Total Probability Theorem

The theorem of total probability can be extended to any number of mutually exclusive events. If the events $A_1, A_2, A_3, \dots, A_k$ are mutually exclusive, then the probability of occurrence of any one of them ($\bigcup_{i=1}^k A_i$) is given by the sum of their probabilities.

$$P(\bigcup_{i=1}^k A_i) = P(A_1 + A_2 + A_3 + \dots, + A_k) = P(A_1) + P(A_2) + P(A_3) + \dots + P(A_k)$$

3) Theorem of Total Probability with Mutually Non-exclusive Events

The probability of occurrence of at least one of the events A and B (which are not necessarily mutually exclusive) is given by

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

The symbol ' $\cap$ ' means joint occurrence or 'and'. Hence, $(A \cap B)$ means the occurrence of the event A and B, whereas ' $\cup$ ' means 'or' i.e., $(A \cup B) \Rightarrow$ the occurrence of either the event A or the event B.

Proof: The occurrence of the event $(A \cup B)$ is analogous to the occurrence of any one of the following three mutually exclusive events:

$(A \cap B^c)$, $(A^c \cap B)$ and $(A \cap B)$. i.e.,

$$P(A + B) = P(A \cup B) = P(A \cap B^c) + (A^c \cap B) + (A \cap B) = P(A \bar{B}) + P(\bar{A} B) + P(AB)$$

In terms of Venn diagram,

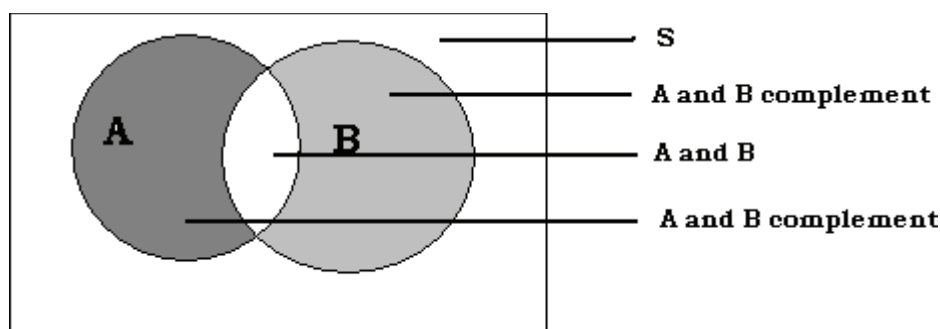


Fig. 25.3: Union and intersection of A & B

Therefore, using the definition of total probability, we get

$$P(A \cup B) = P(A + B) = P(A \cap B^c) + P(A^c \cap B) + P(A \cap B) = P(A \bar{B}) + P(\bar{A} B) + P(AB) \dots\dots\dots[1]$$

Again, the occurrence of A is analogous to the occurrence of any one of the following two mutually exclusive events $P(A \cap B)$ i.e., $P(AB)$ and $P(A \cap B^c)$ i.e., $P(A \bar{B})$. Thus, we get

$$P(A) = P(A \cap B) + P(A \cap B^c) \text{ or } P(A) = P(A \bar{B}) + P(AB) \dots\dots\dots[2]$$

Similarly, for B

$$P(B) = P(B \cap A) + P(B \cap A^c) \text{ or } P(B) = P(AB) + P(\bar{A} B) \dots\dots\dots[3]$$

Using [1], [2], [3] we can derive that

$$P(A \cup B) = P(A + B) = P(A) - P(AB) + P(B) - P(AB) = P(A) + P(B) - P(AB) = P(A) + P(B) - P(A \cap B).$$

The above result could be extended to three events A, B, C, which are not mutually exclusive

$$P(A \cup B \cup C) = P(A) + P(B) + P(C) - P(A \cap B) - P(A \cap C) - P(C \cap B) + P(A \cap B \cap C)$$

We illustrate the situation in Fig. 25.4.

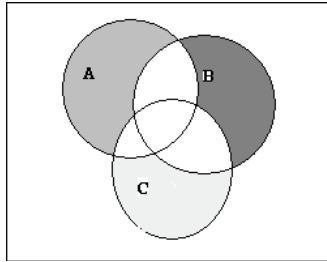


Fig. 25.4: Union and intersection of A, B and C

In this context, we mention that there are two standard results in the theory of probability

- 1) $P(A \cup B) \leq P(A) + P(B)$ Boole's inequality
- 2) $P(A \cap B) \geq P(A) + P(B) - 1$ Bonferroni's inequality

25.5.2 Theorem of Compound Probability

The probability of occurrence of the event A and B is given by the product of the probability of the event A and *conditional probability* of the event B given that A has actually occurred, which is denoted by $P(A|B)$. $P(A|B)$ is given by the ratio of the number of events favourable to the event A and B to the number of events favourable to the event A. Symbolically,

$$P(A \cap B) = P(A) \times P(B|A).$$

Suppose a random experiment has n mutually exclusive, exhaustive and equally likely outcomes among which m_1 , m_2 and m_{12} are favourable to the events A, B and $(A \cap B)$ respectively.

So, $P(A) = m_1/n$;

The number of cases favourable to A as well as B: $P(A \cap B) = m_{12}/n$

Note that event A has actually occurred, and so the occurrence of the event B is limited to only m_1 cases. Thus $P(B|A) = m_{12}/m_1$. Therefore,

$$P(A \cap B) = P(A) \times P(B|A) = \frac{m_1}{n} \times \frac{m_{12}}{m_1} = \frac{m_{12}}{n} = P(A \cap B)$$

This theorem is also known as the multiplication theorem.

25.5.2.1 Deductions from Theorem of Compound Probability

The occurrence of one event, say, B may be associated with the occurrence or non-occurrence of another events say, A. This in turn implies that we can think of B to be composed of two mutually exclusive events $(A \cap B)$ and $(A^c \cap B)$. Applying the theorem of total probability

$$P(B) = P(A \cap B) + P(A^c \cap B)$$

$$= P(A) \times P(B|A) + P(A^c) \times P(B|A^c) \dots \dots \dots [\text{using theorem of compound probability}]$$

1) Extension of Compound Probability Theorem

The above theorem can be extended to include the cases when there are three or more events. Suppose there are three events A, B and C. Then

$$P(A \cap B \cap C) = P(A) \times P(B|A) \times P(C|A \cap B)$$

And soon for more than three events.

Example: Given $P(A) = 1/4$, $P(B) = 2/5$ and $P(A \cup B) = 1/2$ find $P(A|B)$ and $P(B|A)$.

$$P(A \cap B) = P(A) + P(B) - P(A \cup B) = 1/4 + 2/5 - 1/2 = 3/20$$

$$\text{Therefore, } P(A|B) = P(A \cap B) / P(B) = 3/20 \times 5/2 = 3/8 \text{ and } P(B|A) = P(A \cap B) / P(A) = 3/20 \times 4 = 3/5$$

Example: At an examination in three courses A, B and C the following results were obtained

25% of the candidates passed in course A

20% of the candidates passed in course B

35% of the candidates passed in course C

7% of the candidates passed in course A and B

5% of the candidates passed in course A and C

2% of the candidates passed in course B and C

1% of the candidates passed in all the subjects.

Find the probability that a candidate got pass marks in atleast one course.

$$P(A) = 0.25, P(B) = 0.2, P(C) = 0.35, P(A \cap B) = 0.07, P(C \cap B) = 0.02, P(A \cap C) = 0.05 \text{ and } P(A \cap B \cap C) = 0.01.$$

$$\text{Therefore, } P(A \cup B \cup C) = .25 + .2 + .35 - .07 - .05 - .02 + .01 = .67$$

Check Your Progress 2

- 1) If $P(A) = 1/2$, $P(B) = 1/3$, $P(A \cap B) = 1/4$ find $P(A^c)$, $P(A \cup B)$, $P(A|B)$, $P(A^c \cap B)$, $P(A^c \cap B^c)$, $P(A^c \cup B)$. State whether the events A and B are mutually exclusive, equally likely, exhaustive.

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- 2) For three events A, B, C which are not mutually exclusive, prove $P(A \cup B \cup C) = P(A) + P(B) + P(C) - P(A \cap B) - P(A \cap C) - P(C \cap B) + P(A \cap B \cap C)$ using Venn diagram.

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25.6 CONDITIONAL PROBABILITY AND CONCEPT OF INDEPENDENCE

25.6.1 Conditional Probability

From the theorem of compound probability we can get the probability of one event, say, event B conditioned on some other event, say A. As we have discussed earlier, this is symbolically written as $P(B|A)$. From the theory of compound probability, we know that

$$P(A \cap B) = P(A) \times P(B|A)$$

or, $P(B|A) = P(A \cap B)/P(A)$ provided that $P(A) \neq 0$.

Example: Find out the probability of getting the Ace of hearts when one card is drawn from a well-shuffled pack of cards given the fact that the card is red.

Let A denote the event that the card is red and B denote the event that the card is the Ace of hearts. Then we are interested in finding $P(B|A)$. From the theorem of conditional probability

$$P(B|A) = P(A \cap B)/P(A) = (13/52) (1/13) / (26/52) = (1/52) (52/26) = 1/26$$

25.6.2 Concept of Independent Events

Two events A and B are said to be statistically independent if the occurrence of one event is not affected by the occurrence of another event. Similarly, several events are said to be independent, if the occurrence of one event is not affected by the supplementary knowledge of the occurrence of other events. This implies

$$P(B|A) = P(B|A^c) = P(B)$$

Therefore, from the theorem of compound probability, we get

$$P(A \cap B) = P(A) \times P(B|A)$$

$$= P(A) \times P(B)$$

Similarly, for three events that are mutually or statistically independent, we have the following results

$$P(A \cap B \cap C) = P(A) \times P(B) \times P(C) \text{ along with } P(A \cap B) = P(A) \times P(B)$$

$$P(C \cap B) = P(C) \times P(B)$$

$$P(C \cap A) = P(C) \times P(A)$$

For more events A, B, C, D to be mutually independent following should hold:

$$P(A \cap B \cap C \cap D) = P(A) \times P(B) \times P(C) \times P(D) \text{ along with}$$

$$P(A \cap B \cap C) = P(A) \times P(B) \times P(C)$$

$$P(A \cap B \cap D) = P(A) \times P(B) \times P(D)$$

$$P(D \cap B \cap C) = P(D) \times P(B) \times P(C)$$

$$P(A \cap D \cap C) = P(A) \times P(D) \times P(C)$$

$$P(A \cap B) = P(A) \times P(B)$$

$$P(C \cap B) = P(C) \times P(B)$$

$$P(C \cap A) = P(C) \times P(A)$$

And soon.....

Thus, two events are said to be independent if the probability of occurrence of both or all equals the product of their probabilities.

For pairwise independence of events the above should hold for any two of the events.

Deductions from the Concept of Independence

If the events A and B are independent then A^c and B^c are also independent.

Since A and B are independent

$$P(A \cap B) = P(A) \times P(B)$$

$$P(A^c \cap B^c) = P(A \cup B)^c \dots \dots [\text{De Morgan's theorem}]$$

$$= 1 - P(A \cup B)$$

$$= 1 - P(A) - P(B) + P(A \cap B)$$

$$= 1 - P(A) - P(B) + P(A) \times P(B) \quad [\because P(A \cup B) = P(A) + P(B) - P(A \cap B)]$$

$$= \{1 - P(A)\} \{1 - P(B)\}$$

$$= P(A^c) \times P(B^c)$$

Example: Given that $P(A) = 3/8$ and $P(B) = 5/8$ and $P(A \cup B) = 3/4$, find $P(B|A)$ and $P(A|B)$. Are A and B independent?

Using the relationship

$$P(A \cup B) = P(A) + P(B) - P(A \cap B), \text{ we get } P(A \cap B) = 1/4$$

Thus, the given information does not satisfy the equation $P(A \cap B) = P(A) \times P(B)$

Therefore, A and B are not independent.

Example: One urn contains 2 white and 2 red balls and a second urn contains 2 white and 4 red balls

- If one ball is selected from each urn, what is the probability that they will be of the same colour?
- If an urn is selected at random and then a ball is selected at random, what is the probability that it will be a white ball?

Ans.

- Let A denote the event that both the balls drawn from each urn are of the same color. A_1 denotes that they are white and A_2 denotes that they are red. Clearly, A_1 and A_2 are two mutually exclusive events. Applying the theorem of total probability,

$$P(A) = P(A_1) + P(A_2)$$

Here A_1 is a compound event formed by two independent events of drawing a white ball from each urn, Therefore, $P(A_1) = \frac{1}{2} \times \frac{1}{3} = \frac{1}{6}$. Similarly,

$$P(A_2) = \frac{1}{2} \times \frac{2}{3} = \frac{1}{3}$$

$$\text{Hence, } P(A) = \frac{1}{6} + \frac{2}{6} = \frac{1}{2}.$$

- A white ball can be selected in two mutually exclusive ways, when urn 1 is selected and a white ball is drawn from it (denoted by the event A) and when urn 2 is selected and a white ball is drawn from it (denoted by the event B).

$P(A) = P(\text{urn 1 is selected}) \times P(\text{a white ball is drawn})$, because selection of urn 1 and selection of a white ball are mutually independent.

$$P(A) = \frac{1}{2} \times \frac{1}{2} = \frac{1}{4}.$$

Similarly,

$$P(B) = \frac{1}{2} \times \frac{1}{3} = \frac{1}{6}$$

Using the theorem of total probability,

$$P(\text{drawing a white ball from an urn}) = P(A) + P(B) = \frac{5}{12}.$$

Check Your Progress 3

- A salesman has 50% chance of making a sale. If two customers enter the shop, what is the probability that the salesman will make a sale?

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- 2) If the events A and B are independent, then show that A^c, B^c, A and B are pair wise independent.

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- 3) If A and B are mutually exclusive events, then show that $P(A|A \cup B) = P(A)/P(A) + P(B)$.

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- 4) A bag contains 8 red and 3 white balls. Two successive draws of 3 balls are made without replacement. Find the probability that the first drawing will give 3 white balls and the second 3 red balls.

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25.7 BAYES' THEOREM AND ITS APPLICATION

Suppose an event A can occur if and only if one of the mutually exclusive events $B_1, B_2, B_3, \dots, B_n$ occurs. If the unconditional probabilities $P(B_1), P(B_2), P(B_3), \dots, P(B_n)$ are known and the conditional probabilities are $P(A|B_1), P(A|B_2), P(A|B_3), \dots, P(A|B_n)$ and known, then the conditional probability $P(B_i|A)$ could be calculated when A has actually occurred.

$$P(A) = \sum_{i=1}^n P(A \cap B_i) = \sum_{i=1}^n P(B_i) P(A|B_i)$$

$$P(B_i|A) = \frac{P(B_i \cap A)}{P(A)} = \frac{P(A|B_i) \times P(B_i)}{\sum_{i=1}^n P(B_i) P(A|B_i)}, \text{ therefore}$$

$$P(B_i|A) = \frac{P(A|B_i) \times P(B_i)}{\sum_{i=1}^n P(B_i) P(A|B_i)}$$

This is known as Bayes' theorem. This is a very strong result in the theory of probability. An example will illustrate the theorem more vividly.

Example: Two boxes contain respectively 2 red and 2 black balls and 2 red and 4 black balls. One ball is transferred from one box to another and then one ball is selected from the second box. If it turns out to be black, what is the probability that the transferred ball was red?

B_1 : the transferred ball was red.

B_2 : the transferred ball was black.

A: the ball selected from the second box is black. $P(B_1) = \frac{1}{2}$

$$P(B_2) = \frac{1}{2}$$

$$P(A|B_1) = \frac{3}{7}$$

$$P(A|B_2) = \frac{5}{7}$$

$$P(B_1|A) = \frac{P(B_1) \times P(A|B_1)}{P(B_1) \times P(A|B_1) + P(B_2) \times P(A|B_2)}$$

$$= \frac{\frac{1}{2} \times \frac{3}{7}}{\frac{1}{2} \times \frac{3}{7} + \frac{1}{2} \times \frac{5}{7}} = \frac{3}{8}$$

Check Your Progress 4

- 1) In a bulb factory there are three machines a, b and c. They produce 25%, 35% and 40% of total product. Of their output 5, 4 and 2 percent of bulbs respectively are defective. If one bulb is selected at random, what is the probability that it was produced by the machine?

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- 2) We have two coins. The first coin is fair with probability of head = $\frac{1}{2}$, but the second coin is loaded with head. Therefore, probability of getting head in the second coin = $\frac{2}{3}$. Suppose we pick one coin at random, we toss it, and the head shows. Find the probability that we picked the fair coin.

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25.8 MATHEMATICAL EXPECTATIONS

Suppose a random experiment has n mutually exclusive and exhaustive outcomes. Let the variable x take values $x_1, x_2, x_3, \dots, x_n$ with probabilities $p_1, p_2, p_3, \dots, p_n$. Then the mathematical expectation or the expected value of the variable is defined as the weighted sum of the values of the variable, the weights being the probabilities of the values of the variable. Expected value of a variable 'x' is denoted by $E(x)$. Therefore,

$$E(X) = p_1x_1 + p_2x_2 + \dots + p_nx_n$$